

Aaryan Nakhat

+91-9505479453 | aaryan.nakhat@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Founding AI/ML Engineer with 3+ years building production LLM systems end-to-end — fine-tuning and preference-tuning (SFT, DPO), retrieval from vector search to **GraphRAG**, and real-time multi-agent voice infrastructure — with measured production impact (**50% lower on-call latency**, **85% lower operating cost**). Gold-medalist B.Tech in Data Science & AI (**CGPA 9.78/10**). Also ship substantial independent systems from scratch: a primary-source **equity-research & forensic-analysis workbench** for Indian markets and **CricDex**, an open cricket-intelligence platform — both spanning anti-bot data ingestion, deterministic + Bayesian modeling, LLM reasoning, and always-on delivery.

EXPERIENCE

Founding AI/ML Engineer

Sep 2023 – Present

SalesAgents AI

Mohali, India

- **Data & evaluation infrastructure:** Built robust data pipelines for prep, cleaning, transformation, and synthetic-data generation, plus **continuous-evaluation and automated-improvement pipelines** that cut manual effort in the eval-and-improvement loop by **75%**
- **LLM fine-tuning:** **Fine-tuned** multiple small-sized **LLMs** for diverse use cases including conversational AI, bidirectional translation, structured information extraction, topic modeling, action prediction, and custom guardrails
- **Preference alignment:** **Preference-tuned** the fine-tuned LLMs using techniques like **DPO** to better align model outputs with user preferences and improve response quality
- **RAG from scratch:** Iteratively built a **RAG system** starting with **vector-similarity** retrieval (semantic/agentive chunking, re-ranking, query rewriting, HyDE), evolving to **Neo4j** knowledge graphs, and reaching best performance with **GraphRAG** (LightRAG)
- **Voice-agent backend:** Developed and scaled **AI backend systems** on LiveKit — **multi-agent orchestration, tool calling, caching, load balancing, and observability** for production-grade voice agents
- **Real-time optimization:** Optimized real-time voice-agent infrastructure, **reducing on-call latency by 50%** and **cutting operational cost by 85%** through efficient system design and resource utilization

Software Development Intern

Jan 2023 – Sep 2023

ValueLabs

Hyderabad, India

- **Domain LLM in production:** Developed a custom **fine-tuned insurance chat expert**, deployed on a large insurance company's website
- **Guardrails:** Designed **custom guardrails** on top of the fine-tuned LLM **for input and output control**
- **Goal-directed control:** Created an additional **auxiliary classifier** to guide the LLM toward its desired goal at each step

EDUCATION

ICFAI Foundation for Higher Education

Hyderabad, India

Bachelor of Technology in Data Science and Artificial Intelligence; CGPA: 9.78/10

Aug 2019 – Jun 2023

- Achievements: **gold medal** for all-round excellence (best student), **silver medal** for academic excellence, **100% scholarship in all 8 semesters**, won 2 intra-college coding competitions
- Extracurricular activities: vice president of the technical club, part of the university's cricket team
- Relevant Coursework: Probability, Statistics, Data Science, Machine Learning, Deep Learning, Data Visualization, Data Mining, Data Warehousing, Blockchain Technology

PROJECTS

Personal Equity-Research & Forensic Analysis Workbench (Indian Markets) *(code)*

Jun 2026 – Present

- **Primary-source data engine:** Engineered **anti-bot scrapers** (**scrapling** Camoufox browser tier) that pull NSE/BSE filings, XBRL financials, live intraday quotes, F&O participant OI, FII/DII flows, FBIL USD/INR, MCX gold/silver/crude, and AMFI mutual-fund NAVs, all ingested idempotently into a **15+-table DuckDB** store
- **Fundamental, forensic & valuation core:** Multi-year statements, ratios, cash-flow quality and FCFF/FCFE, and a **forensic block** (Altman-Z, Beneish-M, Sloan accruals, Piotroski, promoter-pledge checks); **sector-aware valuation** (P/B-on-ROE, EV/EBITDA, or P/E versus the stock's own history) with a **Monte-Carlo DCF** (20,000 simulations) and a **reverse-DCF** that back-solves the growth the current price implies

- **Ownership & smart-money signals:** A holder-by-holder shareholding breakdown that **classifies each investor** (individual / listed company / mutual fund / foreign fund) and tracks **quarter-over-quarter** changes in who is buying and selling, alongside FII derivatives positioning, **SEBI** insider/promoter trades, promoter pledge, and deduplicated bulk/block deals
- **Idea-discovery engines** (find new ideas, not just analyse named stocks):
 - **Value idea-finder** — ranks the **Nifty-500** on quality, forensic health, and cheapness-versus-its-own-history
 - **Holding-company discount finder** — listed holding companies trading **below the market value of the listed stakes they own**
 - **Marquee-investor tracker** — ~25 well-known investors' disclosed holdings: what each entered, added, trimmed, or exited last quarter
 - **Capex-led small-cap hunter** — small-caps **ramping up capital expenditure**, with distressed / manipulated / heavily-pledged names filtered out
 - **Government policy radar** — mines primary **PIB** releases and maps each scheme to the **sectors and listed companies** likely to benefit
- **Mutual-fund & IPO coverage:** Fund deep-reports across ~14,500 schemes (Sharpe/Sortino/drawdown, **SIP-XIRR** simulation, benchmark alpha/beta/capture, holdings churn) and **IPO** pre-listing notes (fresh-vs-OFS signal, valuation at the band vs peers, subscription and anchor demand → **APPLY/AVOID**), both built only from primary sources
- **Investment thesis & always-on delivery:** An **LLM** reads the quantitative brief and filing PDFs to write a forensic investment thesis (a filing-grounded business overview plus an impact-ranked **growth-triggers scoreboard**), delivered by an **email/Telegram bot**: interactive **PDF** reports and scheduled digests at midday (12:30, live) and end of day (18:00) covering sectoral indices, VIX, FII/DII flows, commodities, top movers, and filing-level event analysis

CricDex: Open Cricket Intelligence Platform (*demo*) (*code*)

May 2026 – Jun 2026

- **Bayesian skill engine:** Rated players with dismissal-aware, opponent-adjusted **Bayesian** models (**JAX/NumPyro**, **ADVI/NUTS**), scoring batters on run-getting and survival and bowlers on economy and strike rate, trained on **Cricsheet** ball-by-ball data held in **DuckDB**
- **Novel cricket metrics:** Designed 10 context-adjusted metrics that measure match impact better than batting average or economy, including **Net Game Impact** (per-ball win-probability added, via an **XGBoost** model), **Wicket Quality**, **Crease Longevity**, **Slow-Start Cost**, etc.
- **Scout & auction simulator:** Built a cross-league **Scout** that finds player look-alikes across 6 T20 leagues using a **Bayesian** skill score, each with an estimated auction price, and a rules-accurate **IPL auction simulator** (**Monte-Carlo**, ~300 runs) supporting real retentions, overseas caps, and second-price bidding

CERTIFICATIONS AND SKILLS

LLM & Agents: LLM fine-tuning (SFT, DPO), RAG → GraphRAG (LightRAG), LangChain, LiveKit multi-agent orchestration, tool-calling, semantic/agentic chunking, re-ranking, query rewriting, HyDE, custom guardrails

Retrieval & Databases: Neo4j knowledge graphs, vector databases, DuckDB, SQL

ML & Serving: PyTorch, JAX/NumPyro, vLLM, SGLang, Docker, Kubernetes, FastAPI

Data & BI: Tableau, Power BI

Cloud & Tools: AWS, GCP, Azure, Linux, Git, VS Code, PyCharm, Jupyter Notebooks

Languages: Python, C, C++, SQL

Certifications (clickable links): Python; Data Science and Machine Learning; Deep Learning; Custom Models, Layers and Loss Functions; Building LLM-Powered Applications; REST APIs; SQL